HOMEWORK 5

Exercise: 1, 6, 7, page 66;
Exercise: 4, 5, 10, 11, 12, 13, page 73-74;

In the following, we fix a field F.

Problem 1. *Let $A \in \mathrm{Mat}_{m\times n}(F)$ be a matrix of row rank 1.*

(1) *Show that there is a matrix $u \in \mathrm{Mat}_{m\times 1}(F)$ and a matrix $v \in \mathrm{Mat}_{1\times n}(F)$ such that $A = uv$. (Here uv means the matrix product of u and v).*
(2) *If $m = n$, show that $A^2 = \mathrm{tr}(A)A$.*

The next problem is a generalization of the above one.

Problem 2. *Let $A \in \mathrm{Mat}_{m\times n}(F)$ be a matrix of rank k. Show that there is a matrix $C \in \mathrm{Mat}_{m\times k}(F), R \in \mathrm{Mat}_{k\times n}$ such that $A = CR$.*

(Hint: use elementary row operations. Here one can take R to be the nonzero rows of the RRE matrix of A. What can you say about C? For an integer p with $p < \mathrm{rank}(A)$, is it possible to find matrices $C_1 \in \mathrm{Mat}_{m\times p}(F)$ and $R_1 \in \mathrm{Mat}_{p\times n}(F)$ such that $A = C_1R_1$? We will go back to this question in future.)

Problem 3. *Let F be a field of characteristic 0. We view elements of F^3 as column vectors. Consider the matrix*

$$A = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & -2 \\ 0 & -1 & 2 \end{bmatrix}$$

and the linear operator $T : F^3 \to F^3$ given by $T(\alpha) = A\alpha$. Compute $\ker(T)$ and $\ker(T^2)$. Moreover, show that $\ker(T^k) = \ker(T)$ for any positive integer k. Here $T^2 = T \circ T$, and $T^k = T \circ T \circ \cdots \circ T$ (k-times).

Problem 4. *Let V be a vector space over F and let $T : V \to V$ be a linear map. Suppose $\ker(T) = \ker(T^2)$. Show that $\ker(T^k) = \ker(T)$ for any $k \geq 1$.*

Problem 5. *We have defined an equivalence relation on the set $\mathrm{Mat}_{m\times n}(F)$ using row equivalence, namely, $A \sim B$ if A and B are row equivalence. Let $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$, the finite field with 3 elements. Consider the set $\mathrm{Mat}_{2\times 3}(\mathbb{F}_3)$. For each equivalence class in $\mathrm{Mat}_{2\times 3}(\mathbb{F}_3)$, find a representative of it. How many equivalence classes are there in $\mathrm{Mat}_{2\times 3}(\mathbb{F}_3)$?*

Given a set X and a positive integer n, recall that the notation X^n denotes the Cartesian product $X \times X \times \cdots \times X$ (n-times) and an element of X^n is of the form $(x_1, \ldots, x_n)$ with $x_i \in X$. Let F be a field and let W be a vector space over F. Then W^n has a natural vector space structure: its addition and scaler multiplication are defined by

$$(\alpha_1, \ldots, \alpha_n) + (\beta_1, \ldots, \beta_n) := (\alpha_1 + \beta_1, \ldots, \alpha_n + \beta_n); c(\alpha_1, \ldots, \alpha_n) : \quad = (c\alpha_1, \ldots, c\alpha_n),$$

for $\alpha_i, \beta_j \in W, c \in F$. Here the notation := has the following meaning: the right side of this notation is the definition of the left side, or its left side is defined in terms of the right side. It should be clear that $\dim_F W^n = n \dim_F W$ if $\dim_F W$ is finite.

Problem 6. *Let V, W be vector spaces over F with $\dim_F V = n$. We don't assume that W has finite dimension. Let $\mathcal{B} = \{\alpha_1, \ldots, \alpha_n\}$ be a fixed (ordered) basis of V.*

(1) *Consider the linear transformation $\theta_{\mathcal{B}} : \mathrm{Hom}_F(V, W) \to W^n$ (the notation $\theta_{\mathcal{B}}$ means that this linear transformation depends on $\mathcal{B}$) defined by*

$$\theta_{\mathcal{B}}(T) = (T(\alpha_1), T(\alpha_2), \ldots, T(\alpha_n)), \forall T \in \mathrm{Hom}_F(V, W).$$

Show that $\theta_{\mathcal{B}}$ is an isomorphism directly (without comparing dimensions).

(2) *Conclude that* $\dim_F \operatorname{Hom}_F(V,W) = n \dim_F W$ *if* $\dim_F W$ *is also finite.*

(Comment: Part (1) is just a restatement of Theorem 1, page 69. A special case of (1) is the isomorphism $\operatorname{Hom}_F(F^n, W) \cong W^n$. An even more special case is $\operatorname{Hom}_F(F, W) = W$.)